

Check-In 3

Calerb Louis Jean, Haoran Huo, Xinqi Tao

Group Name: HeyBene

cjean33@gatech.edu | hhuo30@gatech.edu | taoxinqi23@gatech.edu

1 INTRODUCTION

For this check-in, we are sharing our prototypes and our full evaluation plan. Coming out of the brainstorming and design alternatives phase in Check-In 2, our next step is to actually put the three interaction concepts in front of real users and measure how well each one holds up. This document covers what we built, how we are running the evaluation, and exactly what we are measuring and why. Results and analysis will follow once testing is complete.

2 PROTOTYPES WE BUILT

Rather than going the static route, we built three lightweight functional web prototypes using HTML, CSS, and JavaScript. The reason for that decision was pretty simple: two of our three design alternatives depend on dynamic touch input and one relies on the phone's gyroscope, and you just cannot evaluate those kinds of interactions by tapping through a static mockup. A web prototype lets us hook directly into the DeviceOrientationEvent API for tilt-to-steer and native multi-touch events for the joystick, which gives us a much more realistic testing environment without going way past the scope of a medium-fidelity prototype.

To keep the focus on the interaction mechanics rather than the backend, we used a Wizard of Oz approach: the live camera feed is simulated with a looping video background and telemetry values are hardcoded, so participants can interact with the controls naturally without us needing a live robot in the room.

Here is what each prototype does:

- **Alternative 1 — The Mobile Gamer:** A JavaScript-based dynamic thumbstick that snaps to wherever the user's left thumb first touches the screen, rather than being fixed in one spot. A large pill-shaped gas button sits on the right side. A prominent toggle at the top center switches between Manual and Auto Tracking mode and turns green when auto is active.
- **Alternative 2 — Immersive FPV:** Steering is handled entirely by the phone's gyroscope — tilt left or right to turn the car. A single vertical speed slider sits on

the right edge. With no on-screen directional controls blocking the view, almost the entire screen is the live camera feed.

- **Alternative 3 — One-Handed Casual:** Portrait mode layout with the camera feed on the top half of the screen and a semi-circular swipe pad on the bottom half. Dragging up from the pad accelerates, sliding left or right steers, and releasing triggers an auto-brake. A mode toggle banner is displayed prominently at the top.

3 EVALUATION PLAN

Our evaluation is designed to compare the three prototypes in a way that gives us both concrete numbers and honest user feedback. The core question we are trying to answer is: which interaction approach best bridges the tactile gap we identified in our needfinding, while also making the current drive mode clear at all times?

3.1 Participants and Recruitment

We are recruiting 3 to 4 participants and prioritizing depth over breadth for this round. Our sample will intentionally include a mix of people with prior RC or gaming experience and people with no robotics background at all. That split matters because our Check-In 2 survey showed that users with gamepad experience and users without it bring very different mental models to a phone-based controller, and we want to make sure our final design works for both groups. Participants will be recruited through direct outreach from our team's existing network.

3.2 Evaluation Procedure

We will run moderated, one-on-one sessions using the Think-Aloud Protocol. Each participant will access all three web prototypes on their own smartphone so that their grip and posture are natural rather than forced by an unfamiliar device.

To prevent learning effects from skewing the comparison, we will counterbalance the order the three prototypes are presented across participants meaning each prototype gets tested first, second, and third roughly the same number of times across the full participant group.

For each prototype, every participant will complete the same three tasks in order:

- **Task 1 — Situational Awareness:** Identify the current battery level and confirm that the camera feed is active. This task tests whether the HUD layout surfaces essential information clearly without the user having to search for it.

- **Task 2 — Steering and Control:** "Drive" the robot through an imaginary winding S-curve path using whatever input method that prototype provides, and describe how the controls feel out loud. This is the main task for evaluating the tactile gap we want to see how many times users lose their input point and how confident they feel throughout.
- **Task 3 — Mode Switching:** Find and activate Auto Tracking mode, then confirm out loud how they know the mode has changed. This directly tests our Insight 3 from needfinding: that mode switching needs to be impossible to miss.

3.3 What We Are Measuring

On the quantitative side, we are tracking three things:

- **Task Completion Time:** We will measure how long in seconds it takes each participant to complete Task 3 for each prototype. We expect Alternative 3 to be fastest here since its mode toggle is the largest and most prominent, but we want the data to confirm that rather than just assuming it.
- **Control Slips:** We are defining a control slip as either the user's thumb drifting entirely outside the virtual joystick boundary causing the robot to stop or a wrong gesture while trying to steer. We are being specific about this definition because our prototypes are functional enough to actually detect these events, and having a concrete count gives us something real to compare across the three alternatives rather than just relying on impressions.
- **Post-Task Likert Ratings:** After each prototype, participants will rate three statements on a 7-point scale (1 = Strongly Disagree, 7 = Strongly Agree). The three statements map directly to our three core insights: ease of steering addresses the tactile gap, confidence in finger placement gets at whether the floating joystick actually helps, and clarity of mode status tests our mode-switching insight. The full questionnaire is in the appendix.

On the qualitative side, we will record think-aloud audio during each session and run a short post-test interview at the end. The post-test question is simple: which prototype made you feel most confident controlling the robot without having to look at the screen? After all sessions are done, we will transcribe the recordings and group recurring comments into themes things like input feedback, screen visibility, and mode confusion to surface patterns that the numbers alone might not capture.

3.4 How We Will Analyze It

Once all sessions are complete, we will split the analysis into two parts.

For the quantitative data, we will calculate the mean, median, and standard deviation for the Likert ratings, error counts, and task completion times across all three prototypes. If the data is well-behaved enough, we will run a repeated-measures comparison to check whether the differences between alternatives are meaningful or just noise, though with only 3 to 4 participants we are not expecting to hit statistical significance the descriptive trends will be more useful at this stage.

For the qualitative data, we will go through the think-aloud transcripts and tag recurring moments of hesitation, confusion, or surprise. The goal is to connect specific interaction decisions — like the floating joystick snapping behavior or the swipe pad auto-brake — to the moments where users either clicked with the design or got stuck. That will give us a clear rationale for which features to carry forward into the final high-fidelity prototype and which ones to drop or rethink.

4 REFERENCES

- Müller, Matthias and Koltun, Vladlen (2021). "OpenBot: Turning Smartphones into Robots". In: 2021 IEEE International Conference on Robotics and Automation (ICRA). IEEE, pp. 5003–5009.

A APPENDIX: EVALUATION MATERIALS

A.1 Moderator Script and Task Prompts

Opening script: "Thank you for participating. We are testing three new design concepts for a robot control app. Just to be clear we are testing the design, not you, so there are no wrong answers at all. As you go through each one, please think out loud and tell us what you are looking at, what you are trying to do, and what you expect to happen."

Task prompts (read to participants verbatim for every prototype):

- **Task 1:** "Check the interface and tell me the current battery percentage."
- **Task 2:** "Steer the robot forward and maneuver it through an imaginary winding path. Describe out loud how the controls feel as you go."
- **Task 3:** "Find a way to switch to Auto Tracking mode and tell me out loud how you know the mode has successfully changed."

Closing question: "Looking back at all three, which one made you feel the most confident controlling the robot without having to look at the screen the whole time? Why?"

A.2 Post-Task Questionnaire

Completed after each prototype. Rated 1–7 (1 = Strongly Disagree, 7 = Strongly Agree):

1. It was easy to steer the robot using this layout.
2. I felt confident in my finger placement without needing to constantly look down at the screen.
3. It was immediately obvious which driving mode the robot was in.